drifting field tutorial

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Contents

1 Introduction

Given a dataset of N observations $X = \{x_i\}_{i=1}^N$ from some unknown data distribution $p(x)$, the goal of generative modeling is to learn a generated distribution that closely approximates $p(x)$. Once learned, we can then generate new samples that resembles the data.

A naive approach is to train a neural network f_θ to transform standard Gaussian Noise $\epsilon \sim \mathcal{N}(0, I)$ into generated samples. The model is then optimized by minimizing some loss between a batch of target data and a batch of network outputs $f_\theta(\epsilon)$.

Notably, the network, or generator, f_θ implicitly models the data distribution p by inducing a generated distribution q. In generative modeling, it is useful to think of network outputs as samples from q because it allows us to directly compare and match distributions.

reference implicit and score-based modeling

Drifting Models [6] focus on the evolution of the generated distribution over training time, which follows a Drifting Field. The Drifting Field acts as a velocity field, which approaches 0 as the generated distribution q matches the data distribution p.

In this tutorial, we review the foundations of Drifting Models and unify them under implicit and score-based perspectives.

2 Background

One way to quantify a probability distribution is with its expected value, or mean. The expected value of a continuous random variable x sampled from distribution p [4], denoted by $E_{x\sim p}[x]$, is:

$$\mathbb{E}_{x\sim p}[x] = \int x\, p(x)\, dx = \int_{-\infty}^{\infty} x\, p(x)\, dx \tag{1}$$

or, if x is discrete:

$$\mathbb{E}_{x\sim p}[x] = \sum x\, p(x) \tag{2}$$

In machine learning, instead of summing or integrating over all possible x, which is intractable (i.e., too expensive to compute), we perform a Monte Carlo Estimate:

$$\mathbb{E}_{x\sim p}[x] \approx \frac{1}{n} \sum_{i=1}^{n} x_i \tag{3}$$

where n is the batch size and x_i is the i^{th} sample in the batch. This allows us to approximate the mean without knowing p.

If we define a distribution by its expected value, we can compare distributions and minimize the difference in expected values as a loss. We can take n samples from a data distribution p, and generate m samples from a generated distribution q, to define the loss:

$$\mathcal{L}_{mean} = \mathbb{E}_{y\sim p}[y] - \mathbb{E}_{x\sim q}[x] = \frac{1}{n} \sum_{i=1}^{n} y_i - \frac{1}{m} \sum_{i=1}^{m} x_i \tag{4}$$

As you may suspect, this will perform poorly for more complicated data because even if the averages perfectly match, p doesn't always equal q.

2.1 Moments

Moments [4] are generalized expected values of a probabiltiy distribution. For each integer n, the n^{th} raw (centered at 0) moment of a distribution p, denoted as $\mu_n(p)$, is:

$$\mu_n(p) := E_{x\sim p}[x^n] \tag{5}$$

For distributions with bounded support (e.g., pixel values ranging from 0 to 255, which is bounded), the distribution is uniquely determined by its moments [4]. We can define a new loss by matching moments:

$$L_{moment} = \sum_{n=1}^{\infty} \mathbb{E}_{y\sim p}[y^n] - \mathbb{E}_{x\sim q}[x^n] \tag{6}$$

with $L_{momement} = 0$ implying $p = q$.

The monomial basis $\{x^n\}_{n=0}^{\infty}$ we used to define moments in Equation (5) introduces practical and theoretical limitations.

For large n, the moments exponentially explode or vanish. Additionally, if x is comprised of multiple features (e.g., an image), we have to compute the moments over the joint distribution, or an even larger sequence of mixed moments (e.g., $\mathbb{E}\left[\prod_{i=1}^{d} x_i\right]$).

In unbounded domains, such as latent spaces, the monomial basis even loses its theoretical guarantees (also known as the Moment Problem [3][9][13][10]).

We can use the standard log normal distribution to prove this [10] with some abuse of notation.

Let x be from the standard log normal distribution (which goes from 0 to ∞), or in other words, $x = e^z$, with $z \sim \mathcal{N}(0, 1)$.

The p.d.f of x, or $p(x)$, is:

$$p(x) = \frac{1}{x\sqrt{2\pi}} e^{-\frac{(\log x)^2}{2}} \quad \text{for } x \geq 0 \tag{7}$$

The n^{th} moment $\mu_n(p)$ of the standard log normal distribution is:

$$\begin{aligned}
\mu_n(p) &= E_{x\sim p}[x^n] && \text{Moment Definition} && (8)\\
&= \int_{-\infty}^{\infty} x^n p(x)\,dx && \text{Expectation Definition} && (9)\\
&= \int_{0}^{\infty} x^n p(x)\,dx && 0 \leq x < \infty && (10)\\
&= \int_{0}^{\infty} x^n \frac{1}{x\sqrt{2\pi}} e^{-\frac{(\log x)^2}{2}}\,dx && \text{Apply Equation (7)} && (11)\\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{nu} e^{-\frac{(u^2)}{2}}\,du && u = \log x \text{ substitution} && (12)\\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{nu-\frac{(u^2)}{2}}\,du && \text{Combine exponents} && (13)\\
&= e^{\frac{n^2}{2}} \int_{-\infty}^{\infty} \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{(u-n)^2}{2}}}_{\text{Gaussian p.d.f}}\,du && nu - \frac{u^2}{2} = -\frac{(u-n)^2}{2} + \frac{n^2}{2} && (14)\\
&= e^{\frac{n^2}{2}} && \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(u-n)^2}{2}}\,du = 1 && (15)
\end{aligned}$$

Now consider a family of sine-perturbed distributions p_α whose p.d.f is defined by:

$$p_\alpha(x) := p(x)\,[1 + \alpha \sin(2\pi \log x)]^{\,1} \tag{16}$$

with the amplitude parameter $0 < \alpha < 1$, and $p(x)$ being the standard log normal p.d.f.

[1] The 1 is added to ensure it still is a valid probability distribution

The n^{th} moment of p_α is:

$$
\begin{aligned}
\mu_n(p_\alpha) &= \int_{-\infty}^{\infty} x^n p_\alpha(x)\,dx && \text{Moment Definition} && (17)\\
&= \int_{0}^{\infty} x^n p(x)\,dx + \alpha \int_{0}^{\infty} x^n p(x)\sin(2\pi \log x)\,dx && \text{Split Integral} && (18)\\
&= e^{\frac{n^2}{2}} + \alpha \int_{0}^{\infty} x^n \frac{1}{x\sqrt{2\pi}} e^{-\frac{(\log x)^2}{2}} \sin(2\pi \log x)\,dx && \text{Apply Equation (7)} && (19)\\
&= e^{\frac{n^2}{2}} + \frac{\alpha}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{nu} e^{-\frac{(u)^2}{2}} \sin(2\pi u) du && u = \log x \text{ substitution} && (20)\\
&= e^{\frac{n^2}{2}} + \frac{\alpha}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{nu-\frac{(u^2)}{2}} \sin(2\pi u) du && \text{Combine exponents} && (21)\\
&= e^{\frac{n^2}{2}} + e^{\frac{n^2}{2}} \int_{-\infty}^{\infty} \frac{\alpha}{\sqrt{2\pi}} e^{-\frac{(u-n)^2}{2}} \sin(2\pi u) du && nu - \frac{u^2}{2} = -\frac{(u-n)^2}{2} + \frac{n^2}{2} && (22)\\
&= e^{\frac{n^2}{2}} + e^{\frac{n^2}{2}} \int_{-\infty}^{\infty} \frac{\alpha}{\sqrt{2\pi}} e^{-\frac{(v)^2}{2}} \sin(2\pi (v+n)) dv && v = u - n \text{ substitution} && (23)\\
&= e^{\frac{n^2}{2}} + e^{\frac{n^2}{2}} \int_{-\infty}^{\infty} \underbrace{\frac{\alpha}{\sqrt{2\pi}} e^{-\frac{(v)^2}{2}}}_{\text{Even Function}} \underbrace{\sin(2\pi v)}_{\text{Odd Function}}\, dv && \sin(v + 2\pi n) = \sin(v) && (24)\\
&= e^{\frac{n^2}{2}} + 0 && \text{Odd function over symmetric integral} = 0 && (25)
\end{aligned}
$$

Thus, different distributions can share the same moments.

However, it is important to note that the sequence of moments is not strictly limited to the monomial basis. Moments can be generalized to an arbitrary sequence of functions $F = (f_n)_{n=0}^{\infty}$. We redefine $\mu_n(p)$ to be generalized n^{th} moment of a distribution p:

$$\mu_n(p) := \mathbb{E}_{x\sim p}\left[f_n(x)\right] \tag{26}$$

Under this formulation, a distribution is represented by the sequence of moments $(\mu_n(p))_{n=0}^{\infty}$ depending on the choice of F.

For the sequence F to uniquely determine the distribution p, F must be sufficiently expressive. Formally, we require the sequence of functions F to form a basis that spans the space of all square-integrable functions on the domain $\mathcal{X}$, denoted as $\mathcal{L}^2(\mathcal{X})$:

$$\mathcal{L}^2(\mathcal{X}) := \left\{ g : \mathcal{X} \to \mathbb{R} \;\middle|\; \int |g(x)|^2\,dx < \infty \right\} \tag{27}$$

In other words, if the linear span of F is dense in $\mathcal{L}^2(\mathcal{X})$, it guarantees that F is capable of approximating any well-behaved function on $\mathcal{X}$ to arbitrary precision. Because $L^2(\mathcal{X})$ is infinite-dimensional, F must consequently be an infinite sequence.

Additionally, the mapping:

$$p \mapsto (\mu_n(p))_{n=0}^{\infty} \tag{28}$$

induced by the sequence of functions from F must be injective (i.e., one-to-one) in order to guarantee identifiability.

2.2 Hilbert space

The sequence $F = (f_n)_{n=0}^{\infty}$ that defines the moments of a distribution naturally induces a mapping into an infinite-dimensional vector space. To compare distributions represented as infinte-dimensional vectors we have to embed our domain to the Hilbert space $\mathcal{H}$, which is a vector space with an inner product.

Consequently, the sequence F induces a feature mapping $\phi : \mathcal{X} \to \mathcal{H}$, defined by:

$$\phi(x) = [f_0(x), f_1(x), \dots]^\top \tag{29}$$

The generalized moment, denoted as μ_p, can then be expressed as:

$$\mu_p = \mathbb{E}_{x \sim p}[\phi(x)] \tag{30}$$

Now we can represent any distribution p as a mean embedding (a single vector) μ_p in the Hilbert (or Feature) space.

We define an objective called Maximum Mean Discrepancy [8][12] (MMD) that compares two distributions p and q in the Hilbert space:

$$\text{MMD}^2(p, q) = \|\mu_p - \mu_q\|^2_{\mathcal{H}} \tag{31}$$

We can then break down the MMD into inner products:

$$\begin{aligned}
&\text{MMD}^2(p, q) = \|\mu_p - \mu_q\|^2_{\mathcal{H}} && \text{MMD Definition} && (32)\\
&= \langle \mu_p, \mu_p \rangle_{\mathcal{H}} - 2\langle \mu_p, \mu_q \rangle_{\mathcal{H}} + \langle \mu_q, \mu_q \rangle_{\mathcal{H}} && \text{Square of a Difference} && (33)\\
&= \langle \mathbb{E}_{x \sim p}[\phi(x)], \mathbb{E}_{x \sim p}[\phi(x)] \rangle_{\mathcal{H}} - 2\langle \mathbb{E}_{x \sim p}[\phi(x)], \mathbb{E}_{y \sim q}[\phi(y)] \rangle_{\mathcal{H}} && &&\\
&\quad + \langle \mathbb{E}_{y \sim q}[\phi(y)], \mathbb{E}_{y \sim q}[\phi(y)] \rangle_{\mathcal{H}} && \text{Apply Equation 30} && (34)\\
&= \mathbb{E}_{x, x' \sim p}[\langle \phi(x), \phi(x') \rangle_{\mathcal{H}}] - 2\mathbb{E}_{x \sim p, y \sim q}[\langle \phi(x), \phi(y) \rangle_{\mathcal{H}}] && &&\\
&\quad + \mathbb{E}_{y, y' \sim q}[\langle \phi(y), \phi(y') \rangle_{\mathcal{H}}] && \text{Linearity of } \mathbb{E} && (35)
\end{aligned}$$

Since we can use Monte Carlo Estimation (Equation 3) for the expectations, we will only be concerned with computing the inner products between the feature maps.

3 Kernels

Computing $\phi(\cdot)$ is intractable, and computing the inner product $\langle \phi(\cdot), \phi(\cdot) \rangle_{\mathcal{H}}$ is even more so.

Instead of the feature mapping $\phi(\cdot)$ we can use kernels to apply something known as the kernel trick [14][7][18][20].

A kernel [14], denoted as $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$, is a function that is symmetric:

$$k(x, y) = k(y, x) \tag{36}$$

and its corresponding kernel matrix K, also known as the Gram matrix:

$$K = \begin{bmatrix} k(x_1, y_1) & k(x_1, y_2) & \dots & k(x_1, y_n) \\ k(x_2, y_1) & k(x_2, y_2) & \dots & k(x_2, y_n) \\ \vdots & \vdots & \ddots & \vdots \\ k(x_n, y_1) & k(x_n, y_2) & \dots & k(x_n, y_n) \end{bmatrix} \succeq 0 \tag{37}$$

is positive semi-definite (PSD) for all $x, y \in \mathcal{X}$. This implies the eigenvalues of K are non-negative.

Equivalently, the kernel is symmetric positive semi-definite:

$$\iint k(x, y) f(x) f(y) \, dx \, dy \geq 0 \tag{38}$$

A Mercer kernel [14] is a kernel that is also continuous and defined over a compact (i.e.,bounded) input space.

We can use Mercer's theorem [14][7] to implicitly represent $\phi(\cdot)$.

Suppose you have a Mercer kernel k with a bounded output:

$$\sup_{x,y} k(x,y) < \infty \tag{39}$$

with the operator sup meaning "least upper bound".

Assume the compact operator T_k takes some function $f(x)$ as its input and outputs a new function:

$$T_k f(x) = \int k(x,y) f(y) dy \tag{40}$$

Because the Mercer kernel k is symmetric and continuous, Spectral Theorem [cite] guarantees that T_k can be decomposed into orthonormal eigenfunctions (i.e., eigen vectors that are functions) $\psi_n(x)$ and corresponding eigenvalues λ_n. Applying the operator T_k on $\psi_n(x)$ scales them by its eigenvalue:

$$T_k \psi_n(x) = \lambda_n \psi_n(x) \tag{41}$$

Furthermore, because k is PSD, the resulting eigenvalues are non-negative:

$$\begin{aligned}
&\iint f(x)k(x,y)f(y)\,dx\,dy \geq 0 && k \text{ is PSD} && (42)\\
&= \int f(x) \left(\int k(x,y) f(y)\,dy \right) dx && \text{Separate Double Integral} && (43)\\
&= \int f(x) T_k f(x)\,dx && \text{Apply Equation 39} && (44)\\
&= \int \psi_n(x) T_k \psi_n(x)\,dx && \text{Substitute } f(x) = \psi_n(x) && (45)\\
&= \int \psi_n(x) \lambda_n \psi_n(x)\,dx && \text{Apply Equation 40} && (46)\\
&= \lambda_n \int \psi_n(x) \psi_n(x)\,dx && \text{Factor out } \lambda_n && (47)
\end{aligned}$$

We then conclude with $\lambda_n \int \psi_n(x)\psi_n(x)\,dx \geq 0$, which implies $\lambda_n \geq 0$.

With non-negative eigenvalues λ_n and orthonormal eigenfunctions ψ_n, Mercer's theorem states that the kernel can be decomposed as a series that converges absolutely and uniformly:

$$k(x,y) = \sum_{n=0}^{\infty} \lambda_n \psi_n(x) \psi_n(y) \tag{48}$$

We can now construct a feature mapping $\phi : \mathcal{X} \to \mathcal{H}$:

$$\phi(x) = \left(\sqrt{\lambda_0}\psi_0(x), \sqrt{\lambda_1}\psi_1(x), \dots \right)^{\top} \tag{49}$$

Importantly, proving that the eigenvalues are non-negative allows us to square root them. Now we can use kernels to evaluate the inner product in the Hilbert space:

$$\begin{aligned}
\langle \phi(x), \phi(y) \rangle &= \sum_{n=0}^{\infty} \sqrt{\lambda_n}\psi_n(x) \sqrt{\lambda_n}\psi_n(y) && \text{Inner product definition} && (50)\\
&= \sum_{n=0}^{\infty} \lambda_n \psi_n(x) \psi_n(y) && \sqrt{\lambda_n}\sqrt{\lambda_n} = \lambda_n && (51)\\
&= k(x,y) && \text{Apply Equation 47} && (52)
\end{aligned}$$

This is the kernel trick, which lets us compare compare two samples in an infinite-dimensional feature space, scaling linearly $O(D)$ with sample dimension D.

Back to Equation (26), a sufficiently expressive sequence F for moment matching is:

$$F = (\sqrt{\lambda_n}\psi_n)_{n=0}^{\infty} \quad \text{for } \lambda_n > 0 \tag{53}$$

which forms an orthogonal basis $\{\lambda_n \psi_n\}_{n=0}^{\infty}$ for the function space $\mathcal{L}^2$ [15].

With this, we can finally compare distributions with MMD:

$$\text{MMD}^2(p,q) = \mathbb{E}_{x,x'\sim p}[k(x,x')] - 2\mathbb{E}_{x\sim p, y\sim q}[k(x,y)] + \mathbb{E}_{y,y'\sim q}[k(y,y')] \tag{54}$$

3.1 Reproducing Kernel Hilbert Space

A Reproducing Kernel Hilbert Space (RKHS) [1][16] is a Hilbert space $\mathcal{H}$ of functions $f : \mathcal{X} \to \mathbb{R}$, uniquely defined by a reproducing kernel.

A reproducing kernel [1] is a kernel $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ that follows two additional properties:

1. For every $x \in \mathcal{X}$, the function $k(x,\cdot)$ is an element in RKHS, denoted as $\mathcal{H}_k$. This allows us to implicitly construct an abstract feature map $\phi : \mathcal{X} \to \mathcal{H}_k$:

$$\phi(x) = k(x,\cdot) \tag{55}$$

2. The Reproducing Property [1], which states:

$$f(x) = \langle f, k(x,\cdot)\rangle_{\mathcal{H}_k} \tag{56}$$

   for every $f \in \mathcal{H}_k$.

In RKHS, we can reproduce the evaluation of a function $f(x)$ with the inner product $\langle f, k(x,\cdot)\rangle_{\mathcal{H}_k}$.

When we set f to $k(y,\cdot)$, we can recover the kernel trick:

$$\begin{aligned}
f(x) &= \langle f, k(x,\cdot)\rangle_{\mathcal{H}_k} && \text{Reproducing Property} && (57)\\
&= \langle k(y,\cdot), k(x,\cdot)\rangle_{\mathcal{H}_k} && f = k(y,\cdot) && (58)\\
&= k(y,x) && f(x) = k(y,x) && (59)\\
&= k(x,y) && \text{Symmetric kernel} && (60)
\end{aligned}$$

This allows us to compute the inner product between the implicit features $\phi(x)$ and $\phi(y)$ with $k(x,y)$.

To build intuition on RKHS, it can be helpful to formulate it from Mercer's theorem.

In Equation (49), $\phi(x)$ maps x to a vector in the Hilbert space. In RKHS, we transition from feature vectors to functions. Any function $f \in \mathcal{H}_k$ can be expressed as a linear combination of the basis of $\mathcal{H}$:

$$f(\cdot) = \sum_{n=0}^{\infty} \alpha_n \sqrt{\lambda_n}\psi_n(\cdot) \tag{61}$$

Consider another function $g \in \mathcal{H}_k$:

$$g(\cdot) = \sum_{n=0}^{\infty} \beta_n \sqrt{\lambda_n}\psi_n(\cdot) \tag{62}$$

We can define an inner product $\langle f, g\rangle_{\mathcal{H}_k}$. For simplicity, we ensure the basis $\{\sqrt{\lambda_n}\psi_n\}_{n=0}^{\infty}$ is orthonormal in $\mathcal{H}_k$, by including a $\frac{1}{\lambda_n}$ term.

$$\begin{aligned}
\langle f, g\rangle_{\mathcal{H}_k} &= \sum_{n=0}^{\infty} \frac{1}{\lambda_n}\langle f, g\rangle_{\mathcal{L}^2} && \mathcal{H}_k \text{ inner product definition} && (63)\\
&= \sum_{n=0}^{\infty} \frac{1}{\lambda_n}\langle(\alpha_n\sqrt{\lambda_n}\psi_n), (\beta_n\sqrt{\lambda_n}\psi_n)\rangle_{\mathcal{L}^2} && \text{Apply Equations 61 and 62} && (64)\\
&= \sum_{n=0}^{\infty} \frac{\alpha_n\beta_n\sqrt{\lambda_n}\sqrt{\lambda_n}}{\lambda_n}\langle\psi_n, \psi_n\rangle_{\mathcal{L}^2} && \text{Bilinearity of inner product} && (65)\\
&= \sum_{n=0}^{\infty} \frac{\lambda_n}{\lambda_n}\alpha_n\beta_n(1) && \text{Eigenfunctions are orthonormal in } \mathcal{L}^2 && (66)\\
&= \sum_{n=0}^{\infty} \alpha_n\beta_n && \frac{\lambda_n}{\lambda_n} = 1 && (67)
\end{aligned}$$

Moreover, we can view the vector $\phi(x)$ as a function $k(x, \cdot)$, which can then be expressed in the same format as f and g:

$$k(x, \cdot) = \sum_{n=0}^{\infty}(\sqrt{\lambda_n}\psi_n(x))\sqrt{\lambda_n}\psi_n(\cdot) \tag{68}$$

with $\sqrt{\lambda_n}\psi_n(x)$ being the weight for $k(x, \cdot)$.

The inner product of f and $k(x, \cdot)$ in space $\mathcal{H}_k$ is:

$$\begin{aligned}
\langle f, k(x, \cdot)\rangle_{\mathcal{H}_k} &= \sum_{n=0}^{\infty} \alpha_n\left(\sqrt{\lambda_n}\psi_n(x)\right) && \text{Apply Equation 57} && (69)\\
&= f(x) && \text{Apply Equation 55} && (70)
\end{aligned}$$

Which brings us back to the Reproducing Property.

One of the key differences between Mercer kernels and Reproducing kernels is that the input space less restricted, because we don't need an explicit form of the feature map. You can think of Reproducing kernels as sort of a generalized Mercer kernel.

In RKHS, we can define the mean embedding of any probability distribution μ_p as a function:

$$\mu_p = \mathbb{E}_{x\sim p}\left[k(x, \cdot)\right] \tag{71}$$

This is also known as a kernel mean embedding [16].

We can also define MMD on RKHS:

$$\begin{aligned}
\mathrm{MMD}^2(p, q) &= \|\mu_p - \mu_q\|^2_{\mathcal{H}_k} && \text{MMD in RKHS} && (72)\\
&= \langle\mu_p, \mu_p\rangle_{\mathcal{H}_k} - 2\langle\mu_p, \mu_q\rangle_{\mathcal{H}_k} + \langle\mu_q, \mu_q\rangle_{\mathcal{H}_k} && \text{Square of a Difference} && (73)\\
&= \langle\mathbb{E}_{x\sim p}[k(x, \cdot)], \mathbb{E}_{x'\sim p}[k(x', \cdot)]\rangle_{\mathcal{H}_k} - 2\langle\mathbb{E}_{x\sim p}[k(x, \cdot)], \mathbb{E}_{y\sim q}[k(y, \cdot)]\rangle_{\mathcal{H}_k} \\
&+ \langle\mathbb{E}_{y\sim q}[k(y, \cdot)], \mathbb{E}_{y'\sim q}[k(y', \cdot)]\rangle_{\mathcal{H}_k} && \text{Apply Equation 71} && (74)\\
&= \mathbb{E}_{x,x'\sim p}[\langle k(x, \cdot), k(x', \cdot)\rangle_{\mathcal{H}_k}] - 2\mathbb{E}_{x\sim p, y\sim q}[\langle k(x, \cdot), k(y, \cdot)\rangle_{\mathcal{H}_k}] \\
&+ \mathbb{E}_{y,y'\sim q}[\langle k(y, \cdot), k(y', \cdot)\rangle_{\mathcal{H}_k}] && \text{Linearity of } \mathbb{E} && (75)\\
&= \mathbb{E}_{x,x'\sim p}[k(x, x')] - 2\mathbb{E}_{x\sim p, y\sim q}[k(x, y)] + \mathbb{E}_{y,y'\sim q}[k(y, y')] && \text{Reproducing Property} && (76)
\end{aligned}$$

3.2 Characteristic and Universal Kernels

A reproducing kernel is called Characteristic [17] with respect to the space of probability distributions $\mathcal{P}$ if the mean embedding map:

$$p \mapsto \mu_p = \mathbb{E}_{x \sim p}[k(x, \cdot)] \tag{77}$$

is injective.

Alternatively, a Mercer kernel is Characteristic, if the corresponding sequence of eigenvalues and eigenfunctions $(\sqrt{\lambda_n}\psi_n)_{n=0}^{\infty}$ form a basis in $\mathcal{L}^2$, like in Equation (53). (Note that the richness of the basis makes it a Universal kernel [15][17], which are a subset of Characteristic kernels). For the purpose of this tutorial, we will view Characteristic kernels as Universal to justify matching infinite moments.

In the context of moment matching, Characteristic and Universal kernels guarantee identifiability. This means that if $\mathrm{MMD}^2(p, q) = 0$, $p = q$.

3.3 Proving Universality

One popular family of kernels are Radial Basis Function (RBF) [11][19][5] kernel.

A RBF kernel $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is defined as:

$$k(x, y) = \gamma(\|x - y\|) \tag{78}$$

with $\gamma(\cdot)$ being some scaling function and $\|\cdot\|$ some distance, typically the Euclidean norm.

It is also important to note that RBF kernels are translation-invariant.

A popular RBF kernel that is also Universal is the Gaussian kernel.

Consider a Gaussian kernel, defined as:

$$k(x, y) = e^{-\sigma^2 \|x-y\|_2^2} \tag{79}$$

for all $x, y \in \mathbb{R}^d$, $\sigma > 0$, with $\|\cdot\|_2$ being the Euclidean Norm.

While we can prove that the Gaussian kernel induces a scaled monomial basis which is Universal in bounded domains (more in Appendix A), we can use Bochner's theorem [2] to prove Universality for unbounded domains [15][17].

Bochner's theorem states that a continuous and bounded function $\gamma : \mathcal{X} \to \mathbb{R}^d$ is positive definite on the input domain $\mathcal{X}$ if and only if it is the Fourier transform of a continuous density function[2] (i.e., a function with a finite total area), denoted as f, on $\mathbb{R}^d$:

$$\gamma(x) = \int e^{-ix^\top \omega} f(\omega) d\omega \tag{80}$$

with $x \in \mathbb{R}^d$.

Suppose γ is part of a space of continuous functions that converges to 0, denoted as $\mathcal{C}_0(\mathbb{R}^d)$.

[2]The formal definition is generalized to finite nonnegative Borel measures, which we will omit for simplicity

4 Drifting Models

4.1 Moment Matching Perspective

4.2 Drifing Field

5 Appendix

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